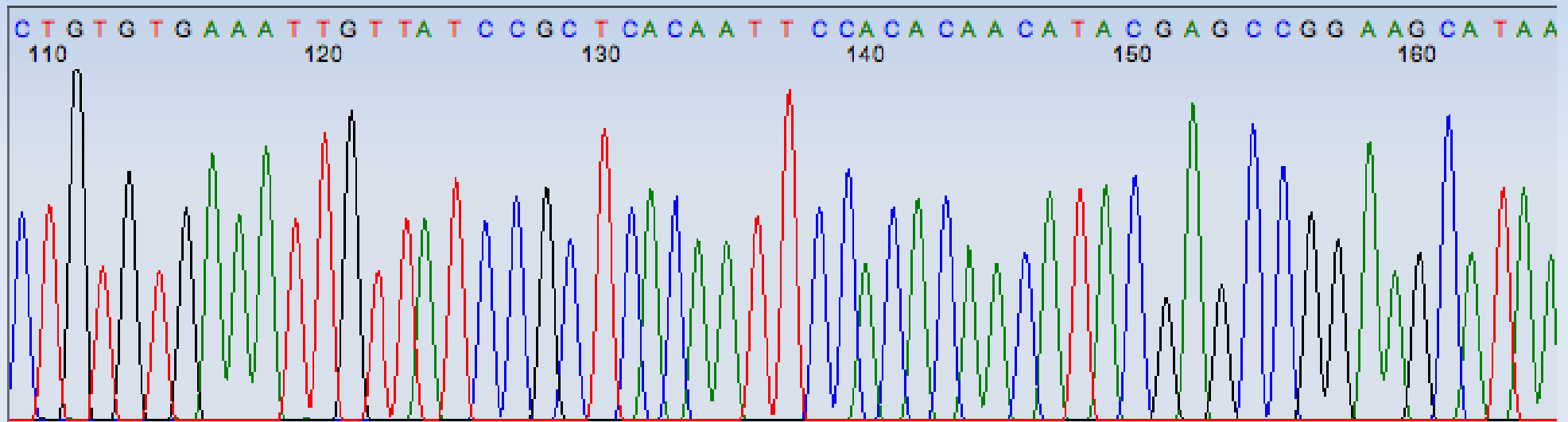


Topic 9.1: Genomes, Sequencing



Sequencing: find the order of nucleotides in a DNA strand

Genomes

Remember, our genome is information about each one of us as individuals – how our cells and bodies are built.

Some questions about eukaryotic genomes:

- Why are they fragmented?
 - Why are genes "everywhere"? Why are they not "organized"?

DNA Transposons - "jumping" genes found by Barbara McClintock in corn
 - Why is a gene "in pieces"? And not all in one place?

Why do we have exons and introns (unlike bacteria)?

Why do the different types of eukaryotes have different numbers and sizes of introns?

Yeast have few introns, plants have many, introns are 25% of our genomes!

1977: introns discovered; 2009: mechanism for creating introns found:

"introner" genomic parasitic DNA sequences that have been creating introns (by copying and pasting themselves into genomes) for billions of years!

Genome Sequencing

Number of base pairs in [human genome](#):

~**3 billion** (haploid – single set of chromosomes)

Cost of sequencing 1st human genome: **\$2.7 bn**

Started ~1990. Completed in stages:

Working draft: Feb 2001

92% complete: April 2003

Telomere-to-telomere [completion](#): **April 2022**

Whose genome was it?

Many people, mainly one anonymous blood donor
from Buffalo, NY

DNA Sequencing

DNA Sequencing is the process of taking a strand of DNA and figuring out the sequence of nucleotides in the strand

Chain termination method ([Sanger](#)):

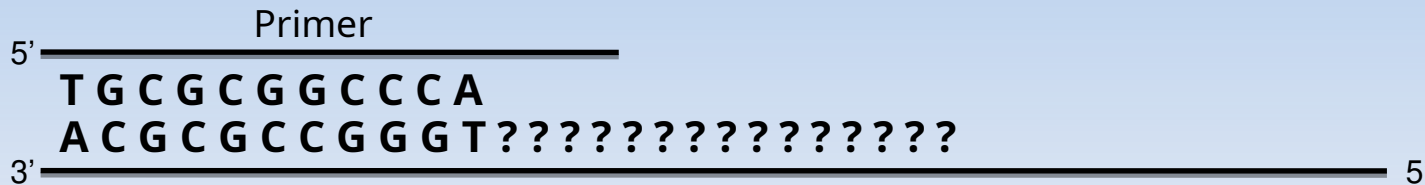
We start with a mostly unknown sequence – suppose we know the 3' end or we add a known sequence at the 3' end:

5' ????????????????????TGGGCCGCGCA 3'

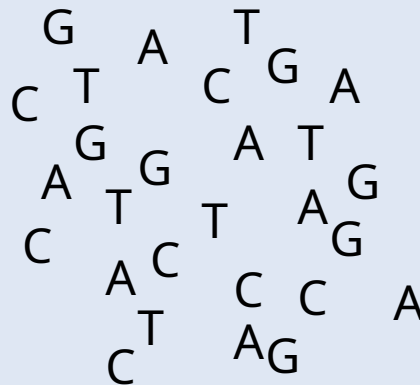
Unknown sequence

Sequencing: Sanger method

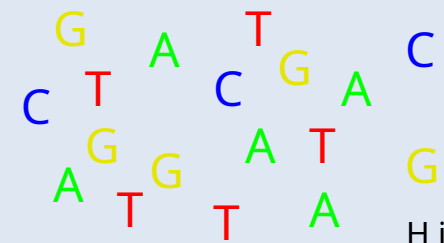
Prime unknown sequence (**primer** binds to vector)



Regular Nucleotides



Dideoxy Nucleotides

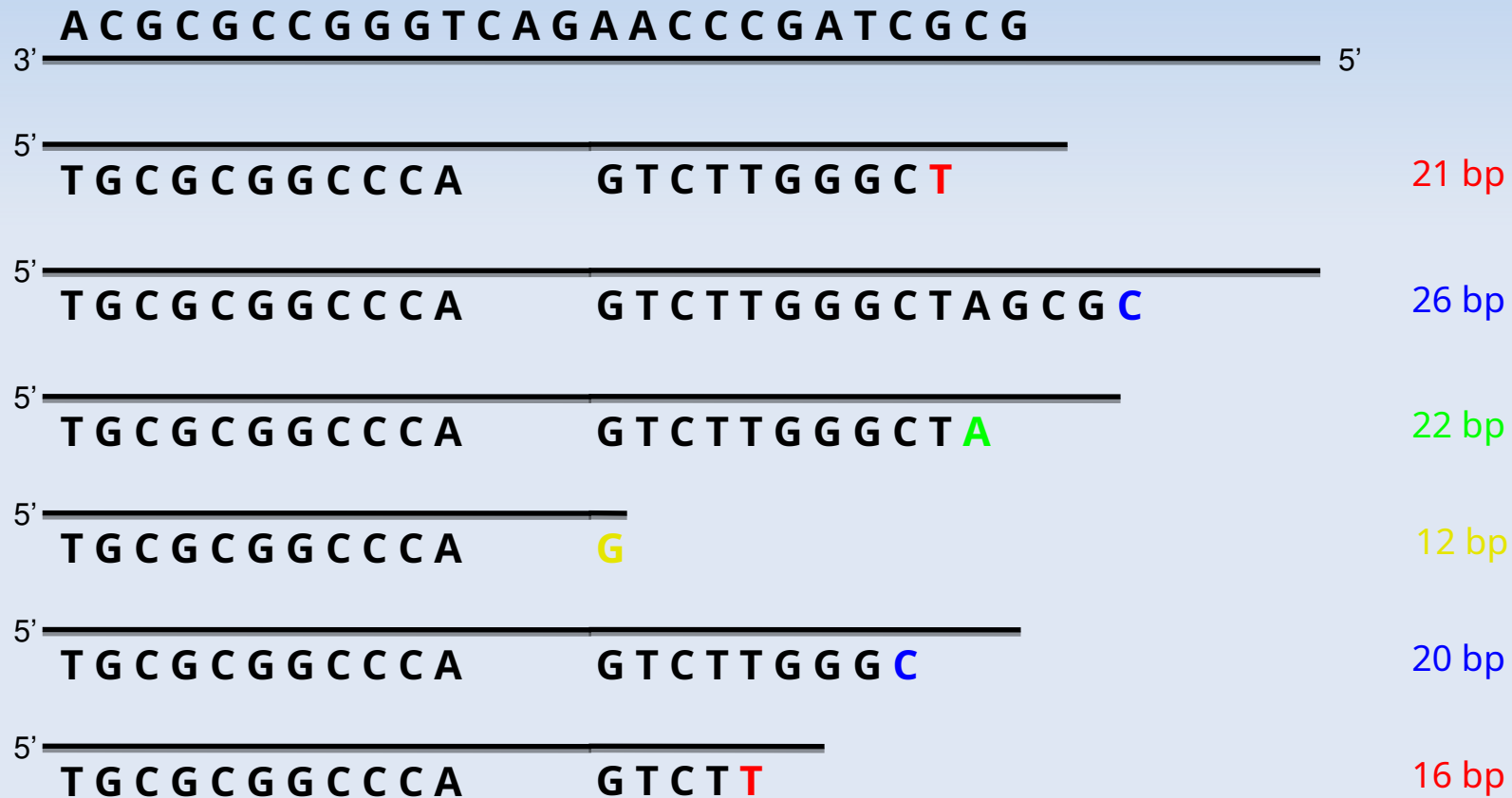


H instead of OH:
cannot form
phosphodiester
bond with next
nucleotide

Add **polymerase** enzyme, "regular" nucleotides and labelled terminating dideoxy nucleotides

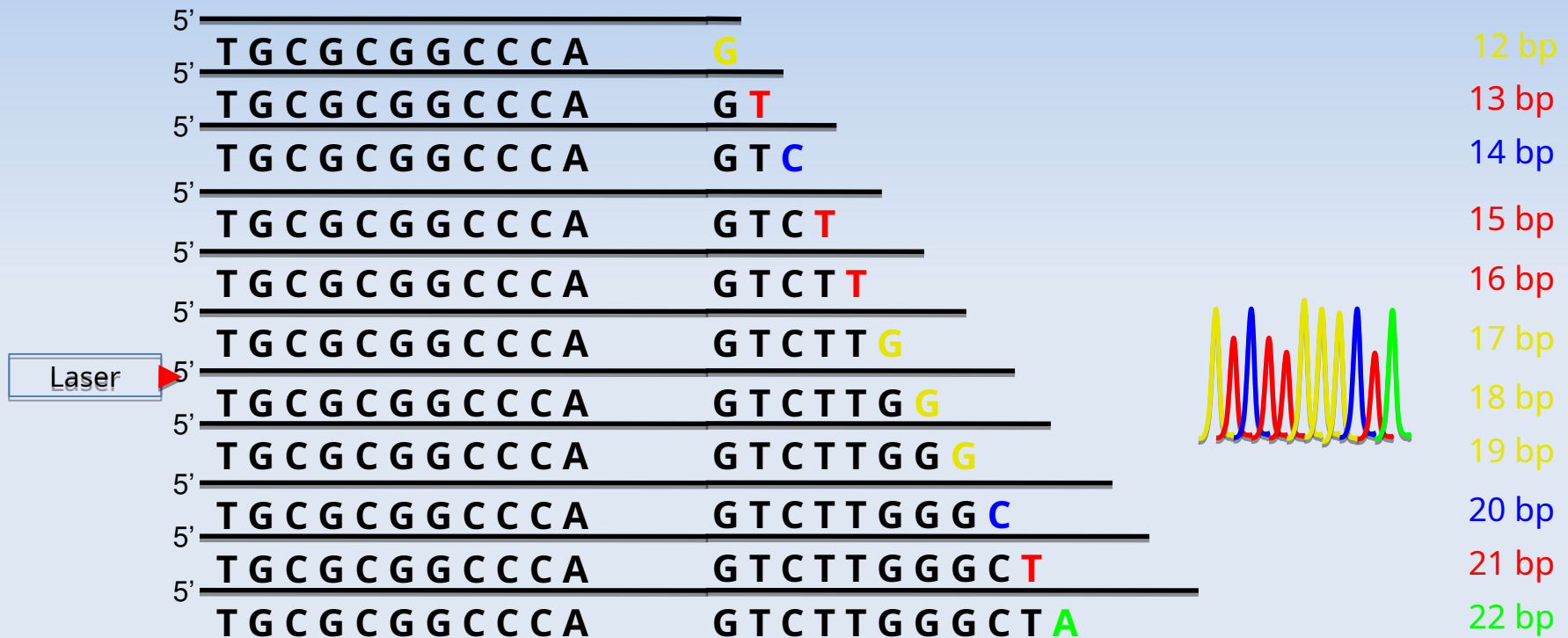
Sequencing: Sanger method

Get many terminated sequences of different lengths

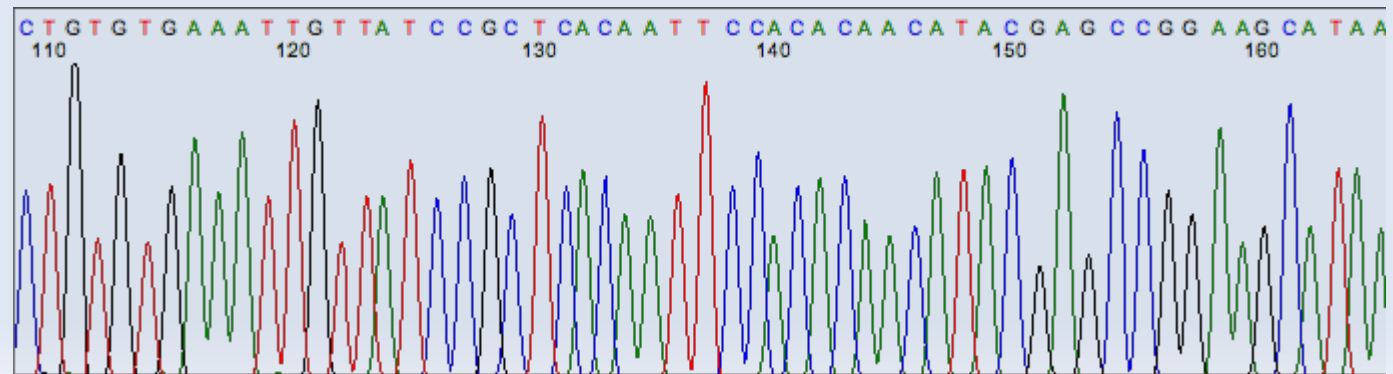


Sequencing: Sanger method

Gel electrophoresis separates products by size; Laser excitation of fluorescent tag → detector "reads" the terminating nucleotides



Gives us a
chromatogram
600-1000 bp

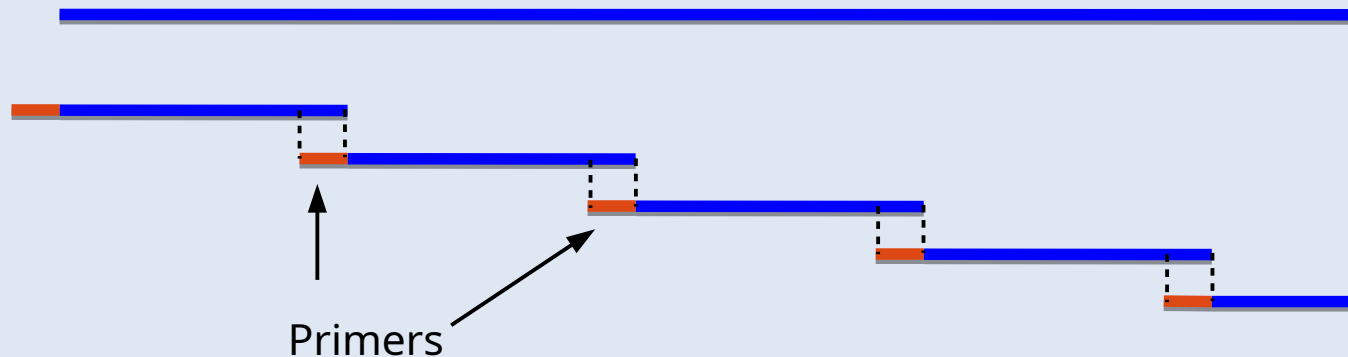


Primer walking

Sanger method works for sequences up to 1 Kbp

For longer sequences we have to use **primer walking** : the tail end of one sequencing run as the primer for the next:
for the next:

Long strand:



First Human Genome

The public human genome project took a slow-but-sure approach of dividing the genome into large fragments (100-200 Kbp).

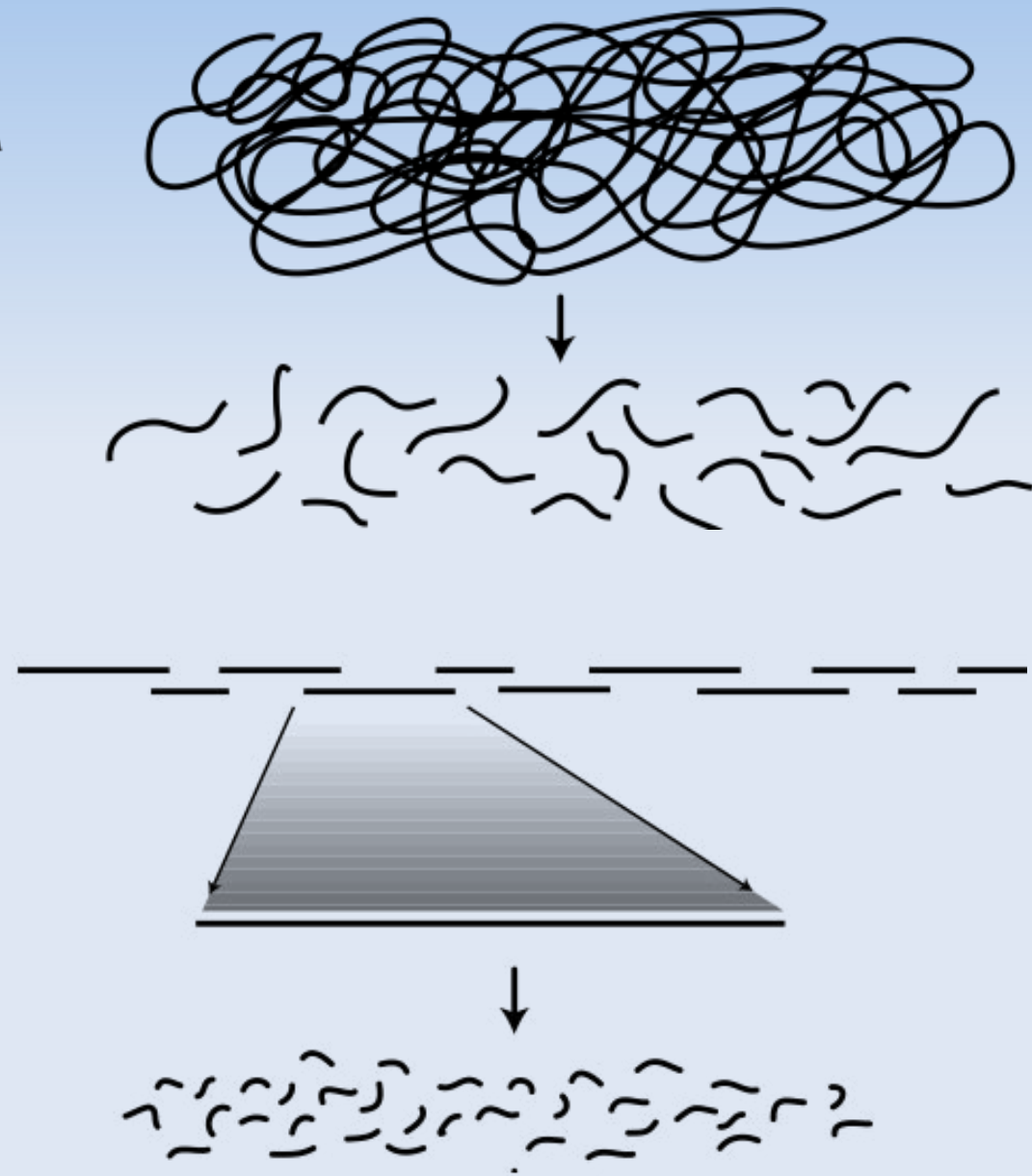
A second cutting step of each BAC contig into 2Kbp fragments allowed sequencing

Genomic DNA

BAC library

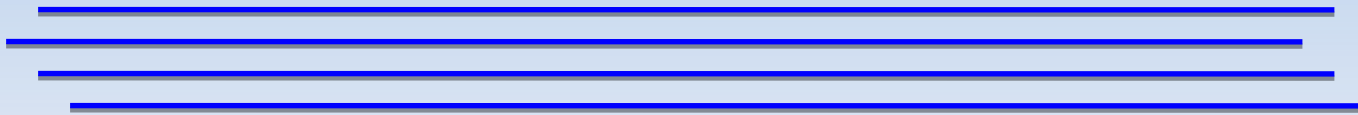
BAC to be sequenced

Shotgun clones



Shotgun Sequencing

Shotgun sequencing starts with many copies of large sequences:



Make enzymatic cuts, sonicate, or hydroshear

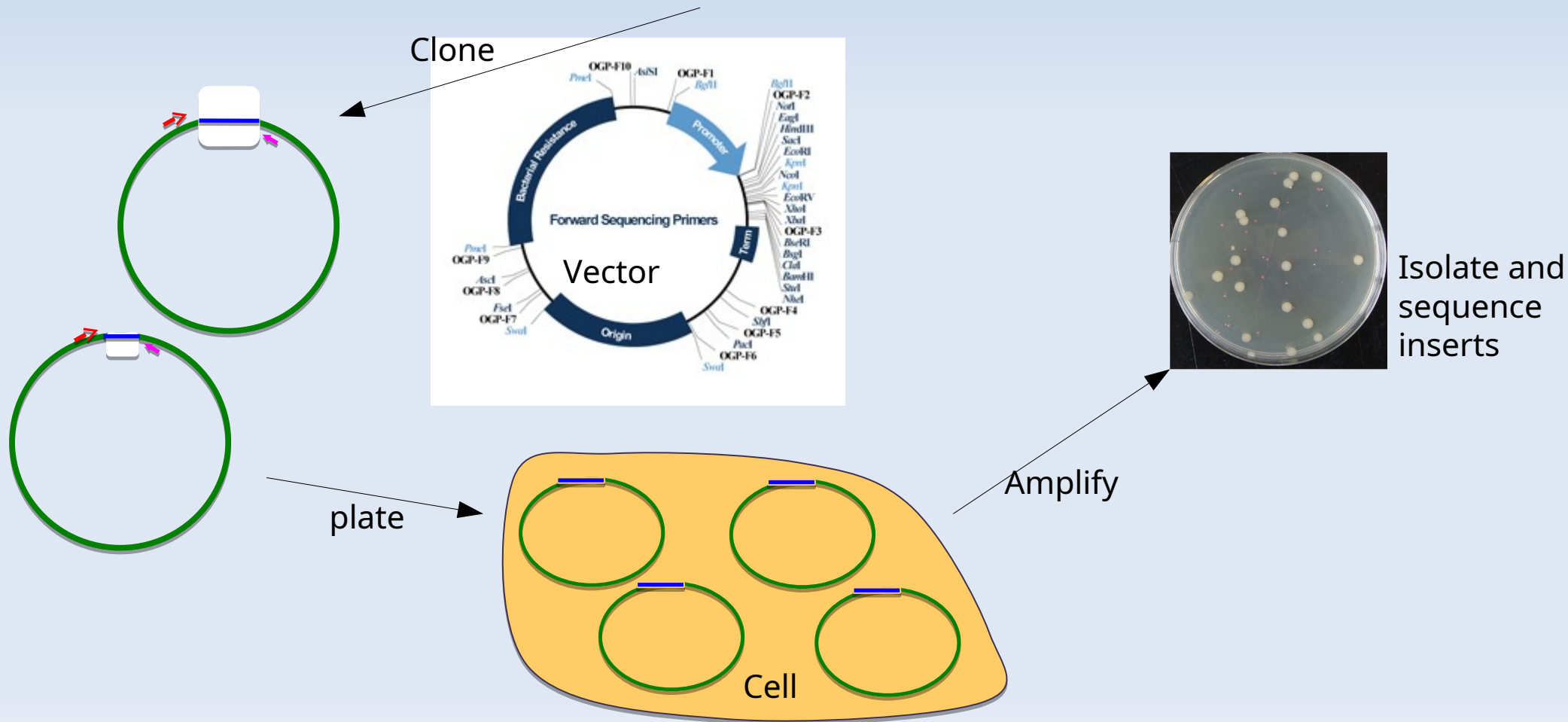


Fragments



Shotgun Sequencing

Clone fragments into a vector, amplify, isolate



Celera Approach

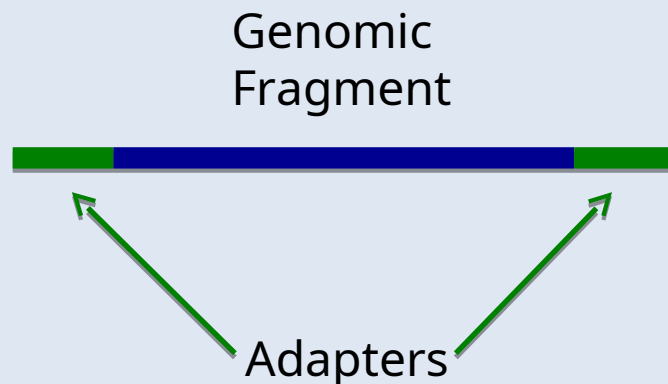
The private (Celera) genome project took the riskier "**whole genome shotgun**" approach

This was faster but required a lot more bioinformatics computations to "assemble" the genome

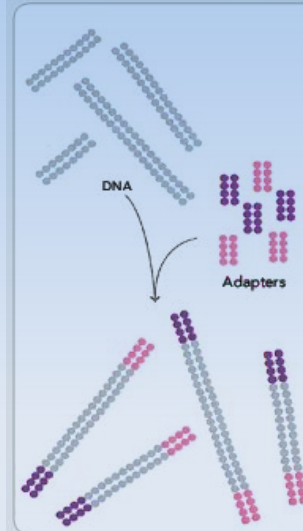
At the time, the computational work was faster than the chemical sequencing part.

Deep or 2nd Generation Sequencing: Solexa sequencing by synthesis

High throughput – many (smaller) fragments, use computational techniques to stitch fragments together.

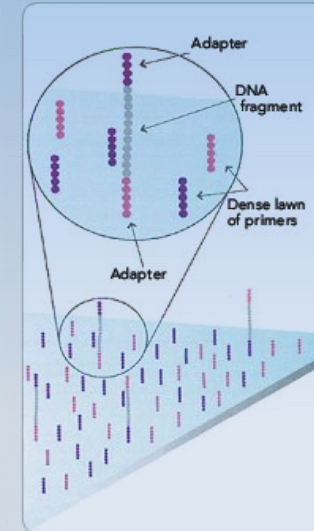


1. PREPARE GENOMIC DNA SAMPLE



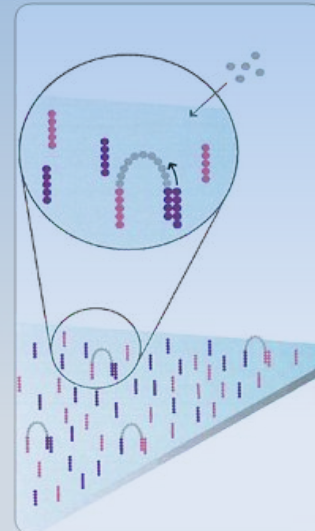
Randomly fragment genomic DNA and ligate adapters to both ends of the fragments.

2. ATTACH DNA TO SURFACE



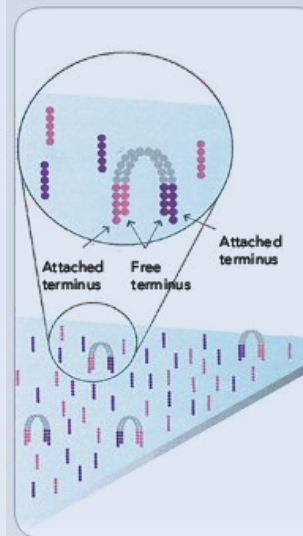
Bind single-stranded fragments randomly to the inside surface of the flow cell channels.

3. BRIDGE AMPLIFICATION



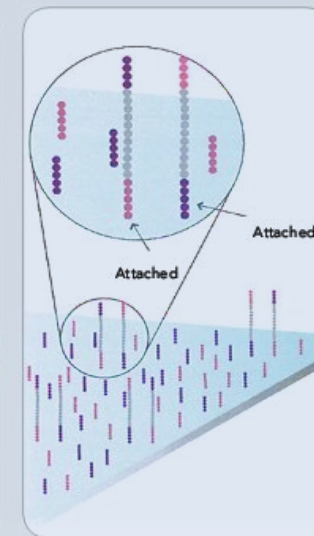
Add unlabeled nucleotides and enzyme to initiate solid-phase bridge amplification.

4. FRAGMENTS BECOME DOUBLE STRANDED



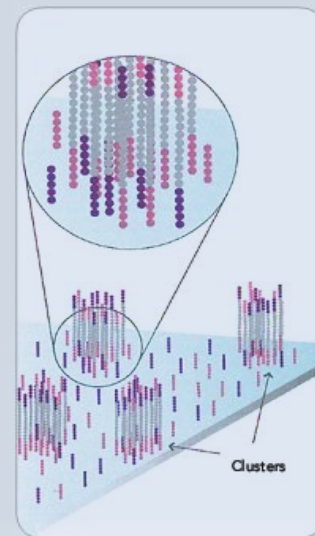
The enzyme incorporates nucleotides to build double-stranded bridges on the solid-phase substrate.

5. DENATURE THE DOUBLE-STRANDED MOLECULES



Denaturation leaves single-stranded templates anchored to the substrate.

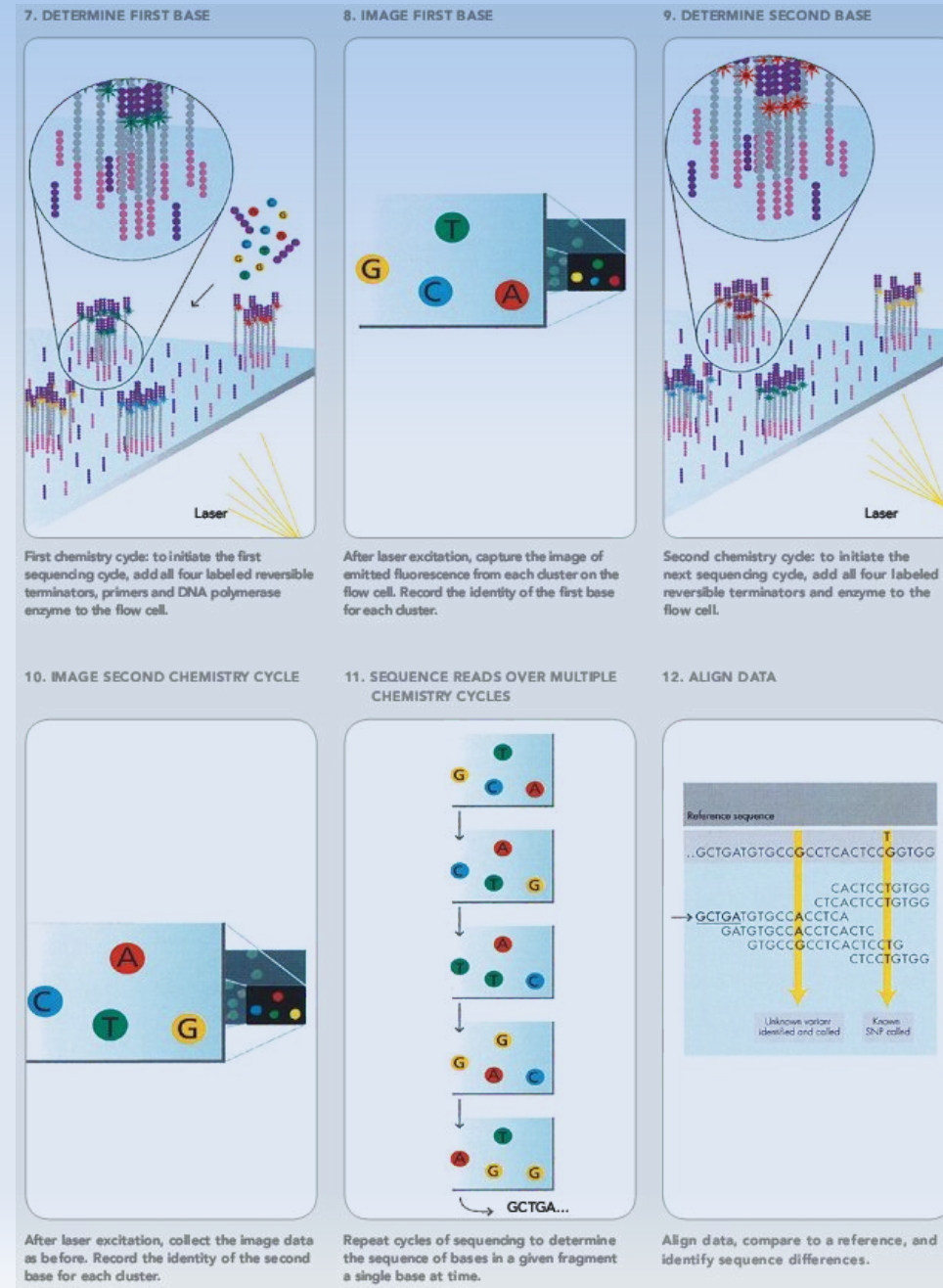
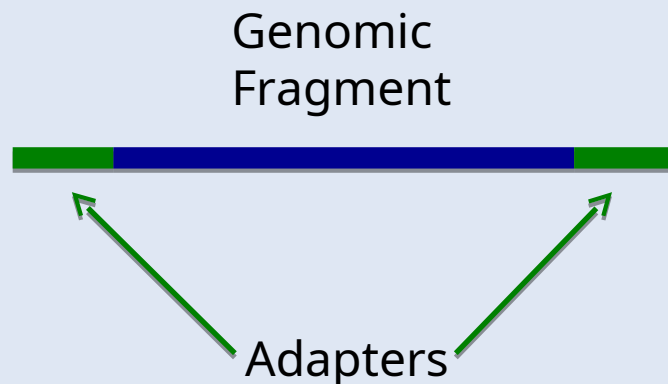
6. COMPLETE AMPLIFICATION



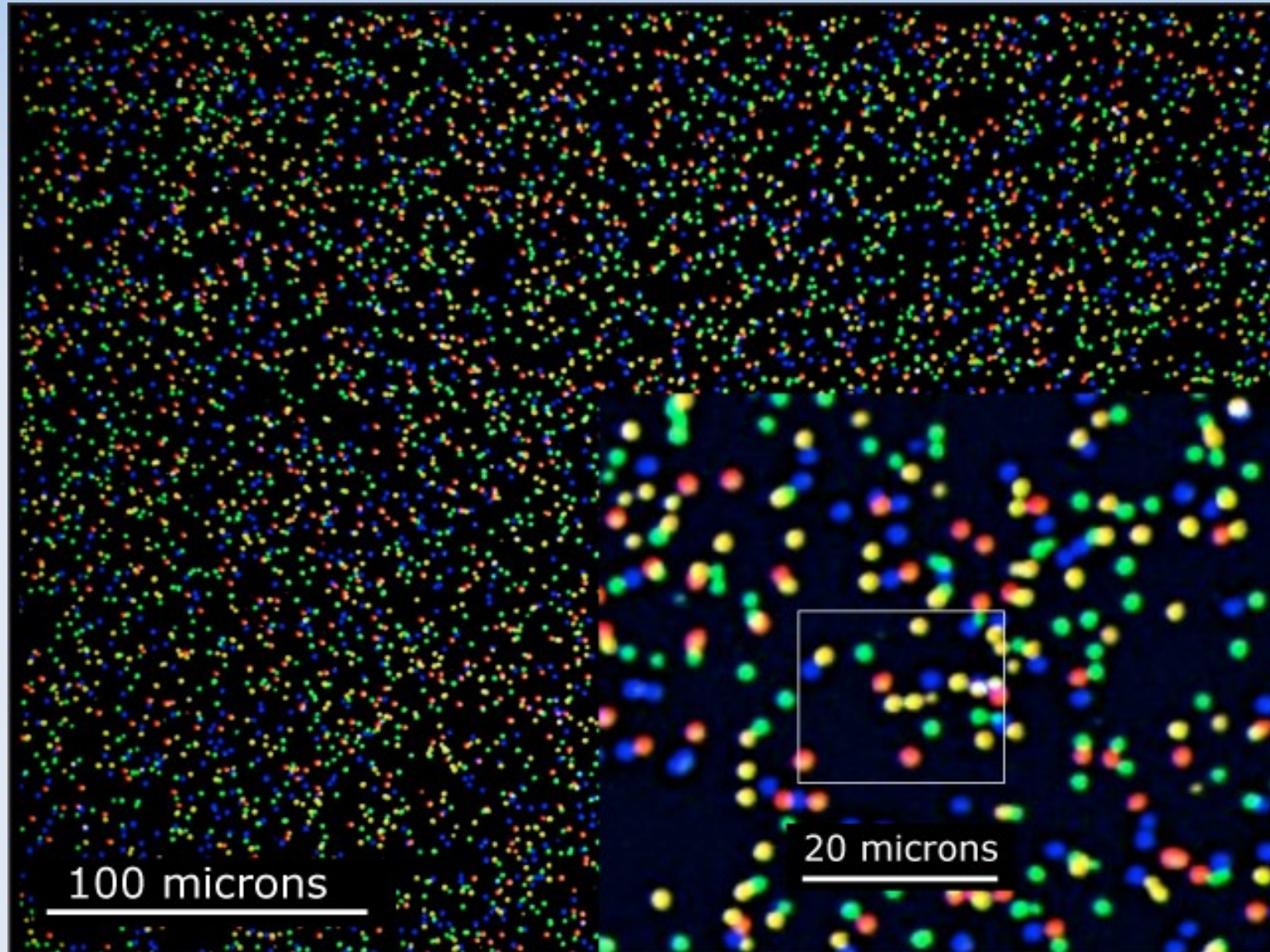
Several million dense clusters of double-stranded DNA are generated in each channel of the flow cell.

2nd Generation Sequencing

High throughput – many (smaller) fragments, use computational techniques to stitch fragments together.



2nd Generation Sequencing



Shotgun Sequencing by Hybridization

(a) Repeats of identical ~200-base DNA fragments are immobilized on a glass surface.

SBH relies on sequential hybridization of different pentamers; one hybridization round with the 3'-ACTAC-5' pentamer is shown.

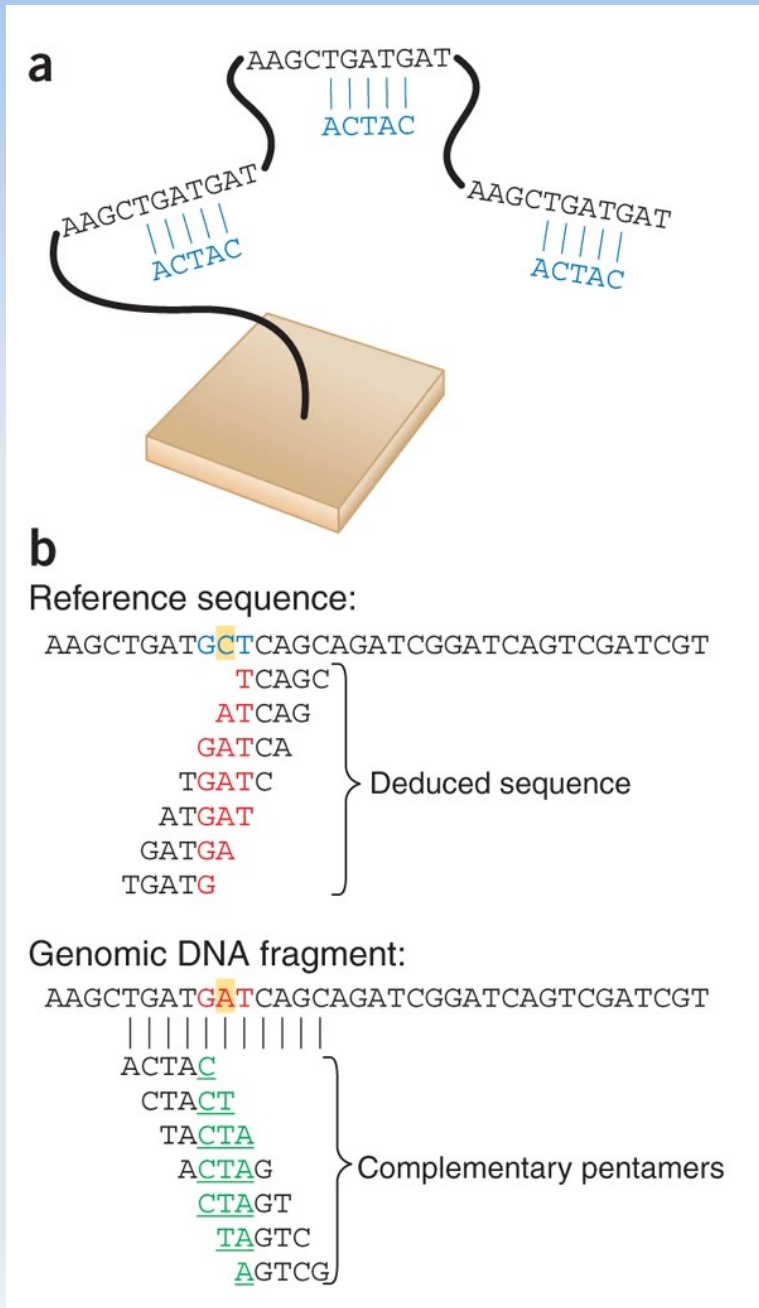
(b) Detection of a point mutation in the trinucleotide sequence GCT (blue in the reference sequence), which shows a change to GAT (red) in a genomic DNA fragment by shotgun-SBH.

The single base change is deduced from five independent complementarity matches (underlined, green) using antiparallel pentamer sequences

The deduced trinucleotide sequence shown in red.

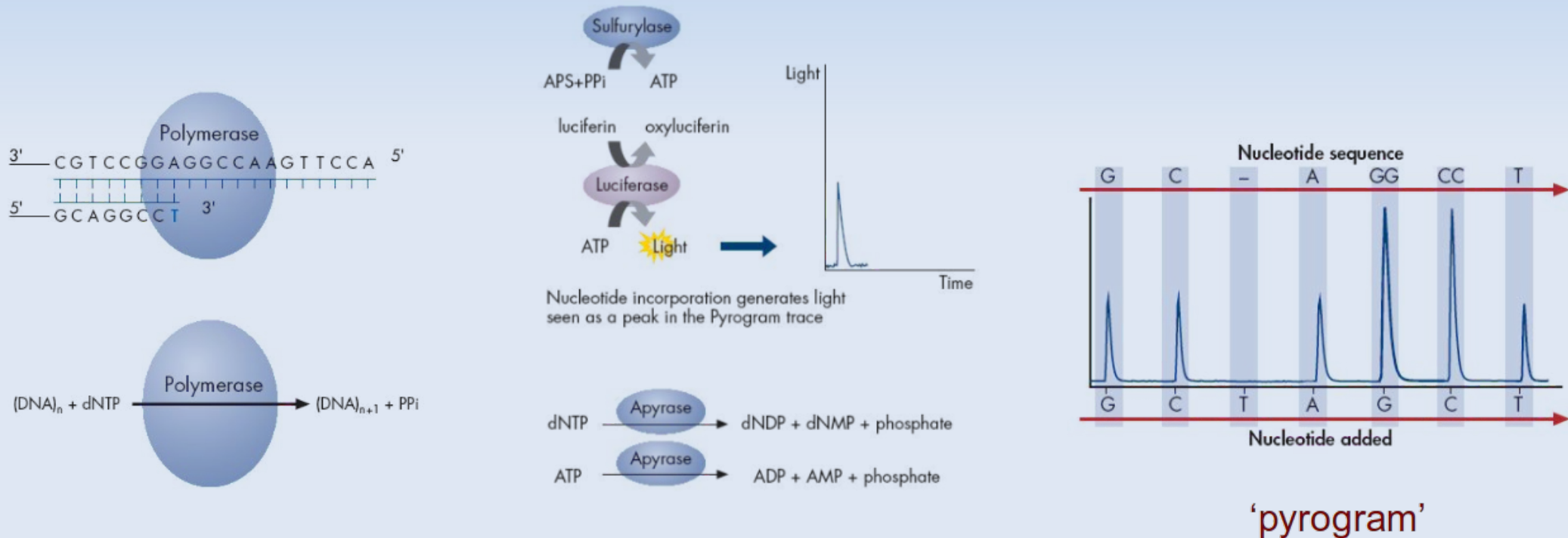
Correctly decoded ~96% of the 48.5-kbp Bacteriophage λ genome and ~80% of the 4.6-Mbp *Escherichia coli* genome in 2008

Worked well for resequencing for the discovery of **single nucleotide polymorphisms**, mutations in coding/noncoding exons



2nd GS: Pyrosequencing

Add nucleotide, Pyrophosphate (PPi) released when nuc. is incorporated into strand:



Detect PPi by converting it to ATP, Luciferase converts it to light signal

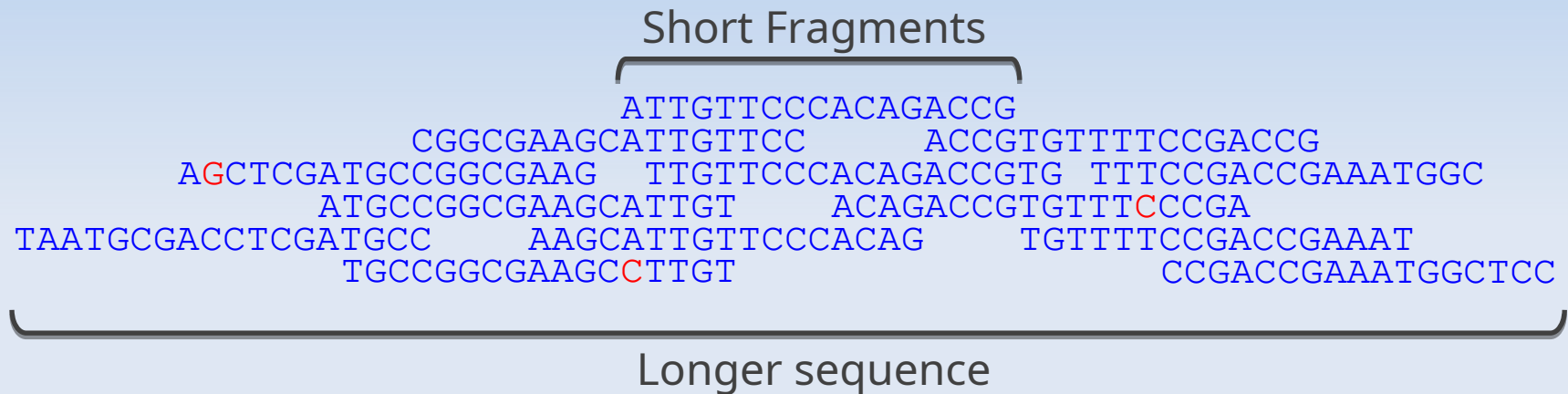
2nd Gen Sequencing

Current sequencers can reliably work with short fragments:

Method	Read Length
Sanger	600-1000 bp
454	400-800 bp
Illumina	75-300 bp
Ion Torrent	~200 bp

Assembly

Assemble fragments into a long sequence



Consensus:

TAATGCGACCTCGATGCCGGCGAAGCATTGTTCCACAGACCGTGTTTTCCGACCGAAATGGCTCC

Coverage: # of reads underlying the consensus

Average coverage: Total bases / consensus length

13 reads * 17 bases per read = **221 bases**

221 bases / 66 base consensus = **3.35-fold coverage**

Assembly

We may luck out

Consensus:

TAATGCGACCTCGATGCCGGCGAAGCATTGTTCCACAGACCGTGTTTTCCGACCGAAATGGCTCC

ATTGTTCCACAGACCG
CGGCGAAGCATTGTTCC ACCGTGTTCGACCG
AGCTCGATGCCGGCGAAG TTGTTCCACAGACCGTG TTTCCGACCGAAATGGC
ATGCCGGCGAAGCATTGT ACAGACCGTGTTTCCGA
TAATGCGACCTCGATGCC AAGCATTGTTCCACAG TGTTCGACCGAAAT
TGCCGGCGAAGCCTTGT CCGACCGAAATGGCTCC

6x coverage

100% identity

Coverage: # of reads underlying the consensus

Assembly

Or not

Consensus:

TAATGCGACCTCGATGCCGGCGAAGCATTGTTCCACAGACCGTGTTTTCCGACCGAAATGGCTCC



ATTGTTCCACAGACCG
CGGCGAAGCATTGTTCC ACCGTGTTTTCCGACCG
AGCTCGATGCCGGCGAAG TTGTTCCACAGACCGTG TTTCCGACCGAAATGGC
ATGCCGGCGAAGCATTGT ACAGACCGTGTTTCCGA
TAATGCGACCTCGATGCC AAGCATTGTTCCACAG TGTTTTCCGACCGAAAT
TGCCGGCGAAGCATTGT CCGACCGAAATGGCTCC

5x coverage
80% identity

Coverage: # of reads underlying the consensus

Assembly

Or not

Consensus:

TAATGCGACCTCGATGCCGGCGAAGCATTGTTCCACAGACCGTGTTTTCCGACCGAAATGGCTCC



ATTGTTCCACAGACCG
CGGCGAAGCATTGTTCC ACCGTGTTTTCCGACCG
AGCTCGATGCCGGCGAAG TTGTTCCACAGACCGTG TTTCCGACCGAAATGGC
ATGCCGGCGAAGCATTGT ACAGACCGTGTTTCCGA
TAATGCGACCTCGATGCC AAGCATTGTTCCACAG TGTTTTCCGACCGAAAT
TGCCGGCGAAGCCTTGT CCGACCGAAATGGCTCC

2x coverage
50% identity

Coverage: # of reads underlying the consensus

Assembly

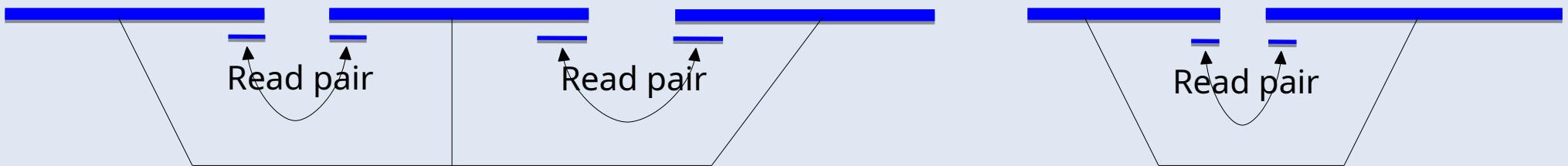
Sequence Reads



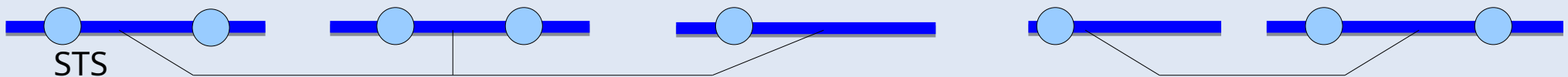
Sequence Contigs: - contiguous segments completely covered by reads



Scaffolds: like contigs but with missing parts - use read pairs and PCR + sequencing to fill in



Mapped Scaffolds

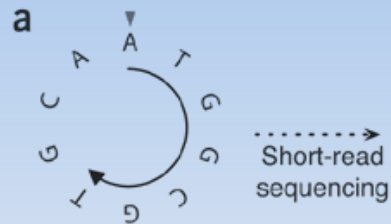


Genome Map

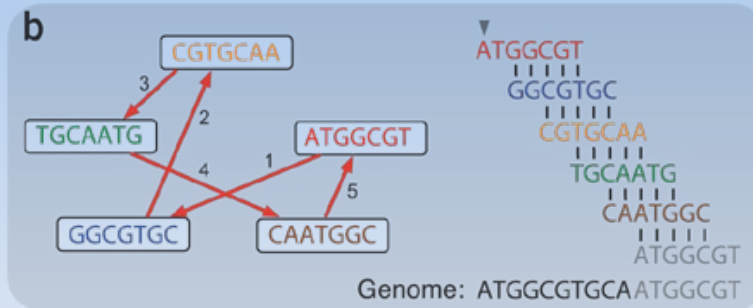


Graphs and Genome Assembly

(a) An example small circular genome.

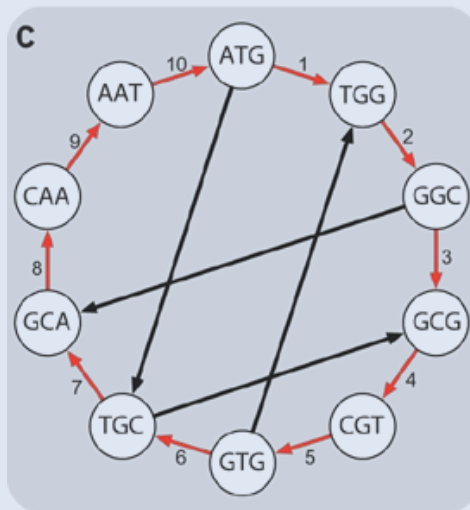


(b) Traditional sequencing algorithms: reads are nodes in a graph, edges are alignments between reads. A Hamiltonian cycle reconstructs the circular genome.



Vertices are k -mers
Edges are pairwise alignments

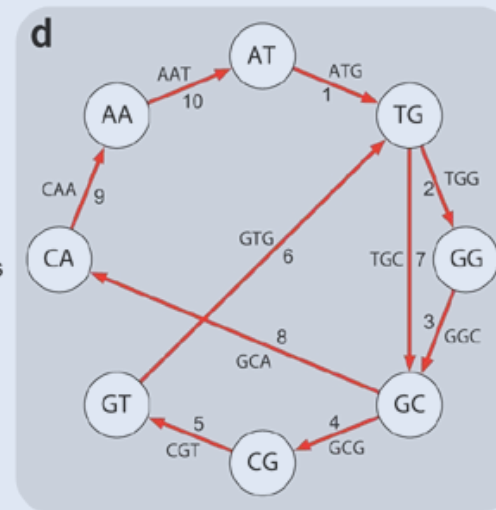
Vertices are $(k-1)$ -mers
Edges are k -mers



Hamiltonian cycle
Visit each vertex once
(harder to solve)



Genome: ATGGCGTGAATGGCGT



Eulerian cycle
Visit each edge once
(easier to solve)

(c) An alternative technique splits reads into k -mers. A Hamiltonian cycle - red - reconstructs the genome by forming an alignment: each successive k -mer is shifted by one position.

(d) Modern short-read assembly algorithms construct a de Bruijn graph. An **Eulerian** cycle reconstructs the genome sequence.

3rd Gen Genome Sequencing

Single DNA molecule sequencing:

- Pacific Biosciences
- Oxford Nanopore
- Quantapore
- Stratos

Main idea is to make longer reads – 10,000 to 100,000 bp
reads → fewer reads needed for assembly →
computationally simpler and faster

3rd Gen Genome Sequencing

PacBio SMRT seq

DNA passes thru
polymerase in an
illuminated volume



Raw output is fluorescent signal
of the nucleotide incorporation,
specific to each nucleotide

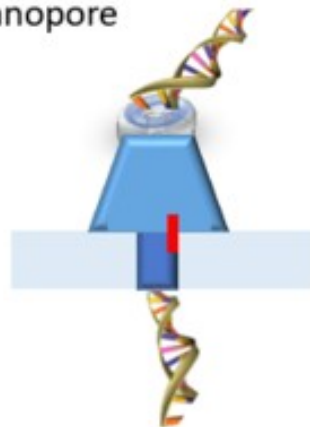


A,C,T,G have known pulse
durations, which are used to
infer methylated nucleotides



Oxford Nanopore

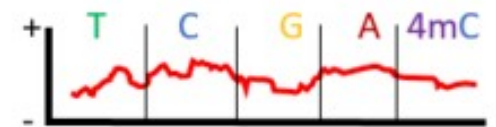
DNA passes thru
nanopore



Raw output is electrical signal
caused by nucleotide blocking
ion flow in nanopore



Each nucleotide has a specific
electric "signature"



Genome Sequencing

Number of base pairs in human genome:

~**3 billion** (haploid – single set of chromosomes)

Current cost of sequencing a human genome

\$1000

Time needed to sequence a human genome in 2022:

1-3 days

Currently: 1000s of human
genomes sequenced so far



Genome Projects

Currently there are ~500,000 organisms with genomes sequenced or being sequenced.

Mostly – ~400,000 – bacteria

~20,000 viruses

~50,000 Eukarya

~200 animals

<https://gold.jgi.doe.gov/index>

Comparative Genomics

By comparing genomes of related animals, we may be able to see what sequences are conserved

5-10% of our genome is conserved cf. other animals

Remember: <1% of our genome is "coding DNA"
(exons that code for protein)

→ a LOT of non-coding functional sequence
information for chromosome packaging, segregation,
and replication; non-coding RNA, and gene regulation

Viewing Genomes

Sites that allow us to view various genomes:

- NCBI: [Genome Data Viewer](#)
- UC Santa Cruz [Genome Browser](#)
- JBrowse <http://jbrowse.org/>
- Ensemble's [Genomes](#)

The free Integrative Genomics Viewer ([IGV](#)) program allows us to access the same data but runs as a standalone program.

Other web sites:

The [gnomAD](#) web site allows us to look at variations in 120,000 genomes.

The [Galaxy](#) web site has one of the largest collection of algorithms for analyzing genomes

Data in Genome Browsers

Besides the actual genomic sequences themselves, these genome browsers usually also have

- RefSeq genes (mRNA and protein)
- Ensembl genes
- Expressed sequence tags
- Gene predictions
- SNPs from a database like dbSNP
- Non-coding functional elements as found by projects like ENCoDE

ENCoDE Project

ENCyclopedia of DNA Elements: Goal is to identify all functional elements in human, mouse genomes

Encode [portal](#):

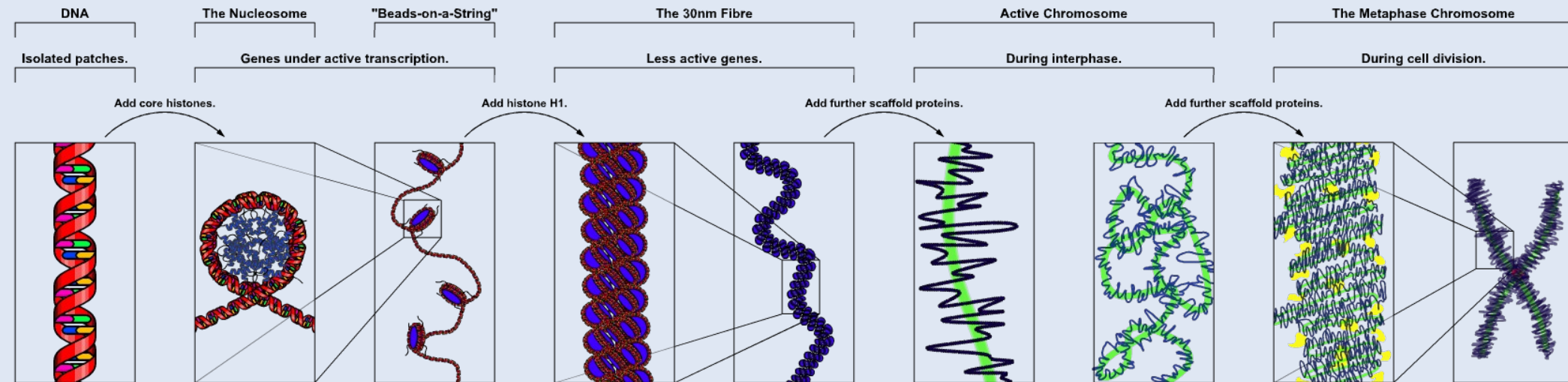
- integrative-level annotations, including a registry of candidate cis-regulatory elements
- ground-level annotations derived directly from experimental data

ENCoDE Project Phase II

ENCyclopedia of DNA Elements: Identify all functional elements in human genome

Phase 4 underway now:

- 80% of human genome participates in biochemical RNA or chromatin (30 nm) process



- 95% of genome lies within 8 kb of a DNA-protein interaction.

Single Nucleotide Polymorphism SNP

An SNP is a confirmed substitution, insertion, or deletion in single nucleotide in a gene (compared to a reference genome)

There are 3-5 million human SN variants

A single individual may have ~60 variants not seen in either parent – mostly biologically silent.

SNPs and Haplotypes

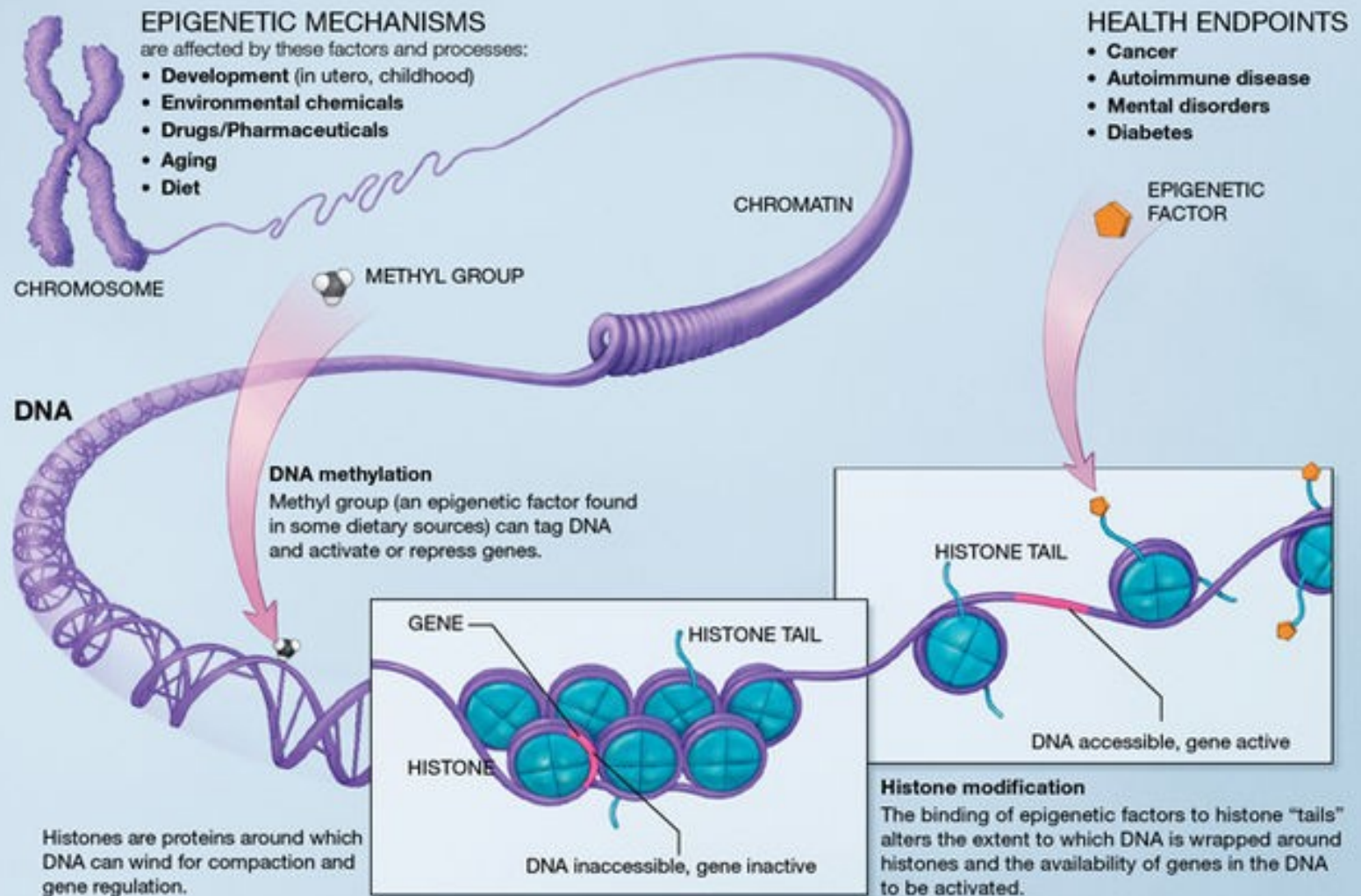
Haplotype – a group of genes that an individual inherits as a unit from a parent; can also refer to a set of SNPs that are inherited together.

A **Haplogroup** is a set of SNPs that represent a clade or group of people with a common ancestor.

- The HapMap project <https://hapmap.ncbi.nlm.nih.gov/>
- 1000 Genomes project – has 2535 human genomes from 26 groups world wide <http://www.1000genomes.org/>

Epigenetics

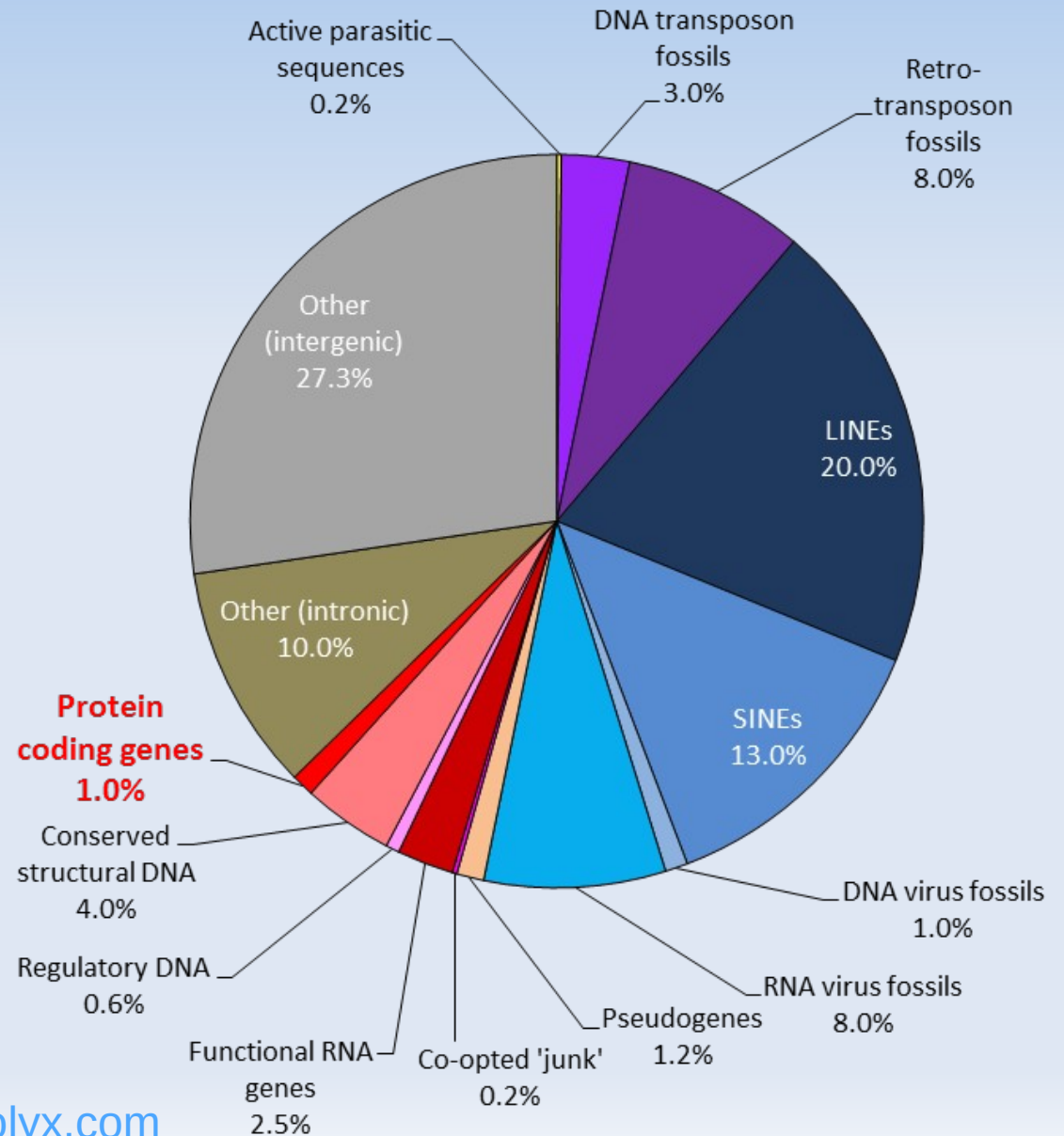
Epigenetics: study of variations in inherited traits caused by environmental factors → 2nd Genomic code



Can get genome wide epigenetic information. e.g. Bisulfite treatment of DNA: converts cytosine to Uracil; methylcytosine is not affected

What's in our genome?

A lot of what we thought was "junk" but only now starting to understand!

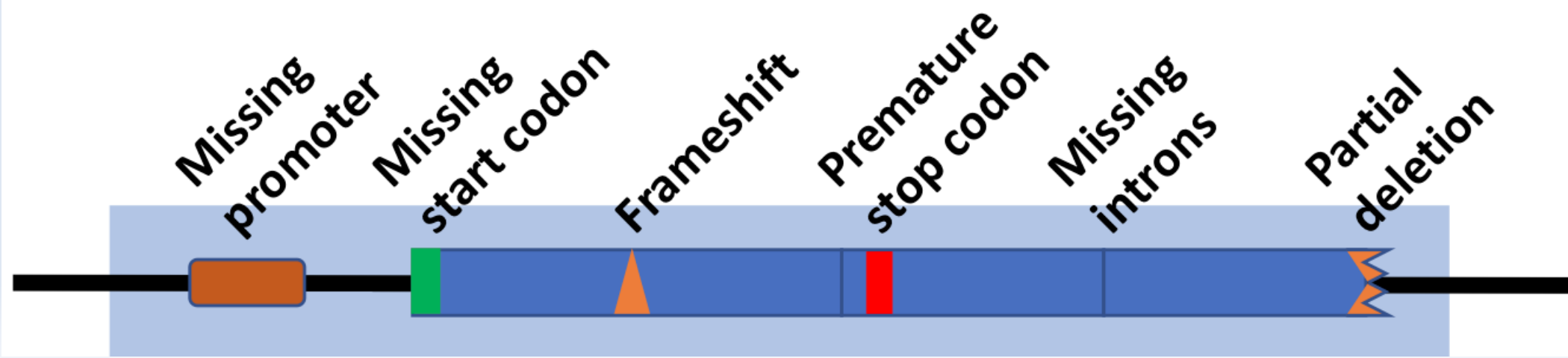


Pseudogenes

About the same fraction of space in our genome that protein coding genes take up is also taken up by **pseudogenes**

→ sequences similar to either some of our own other genes (by gene duplication) or to genes found in other species but have become inactivated by mutations in our genome.

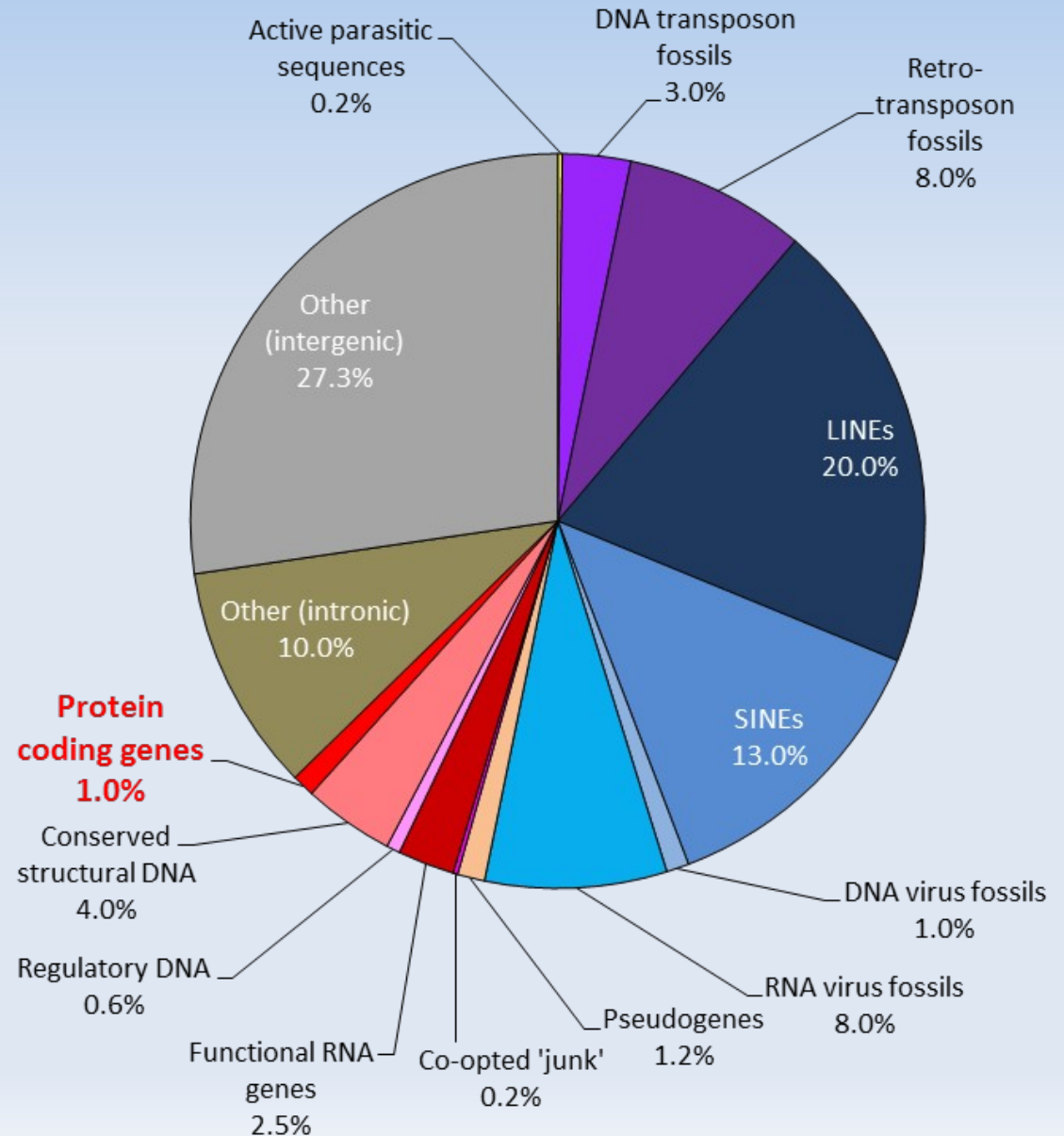
Common defects of pseudogenes:



Introns and intergenic sequences

Introns – the space between exons – may make up 20 times what exons take up

Intergenic sequences – the space between genes with unknown function – may make up another 20 times as much as exons



Transposable Elements

About half of the human genome is taken up by transposable elements or "jumping genes"

Transposable elements are genetic sequences that can change position within our genome. They include

Retrotransposons ~40%

DNA transposons ~3%

Retrotransposons

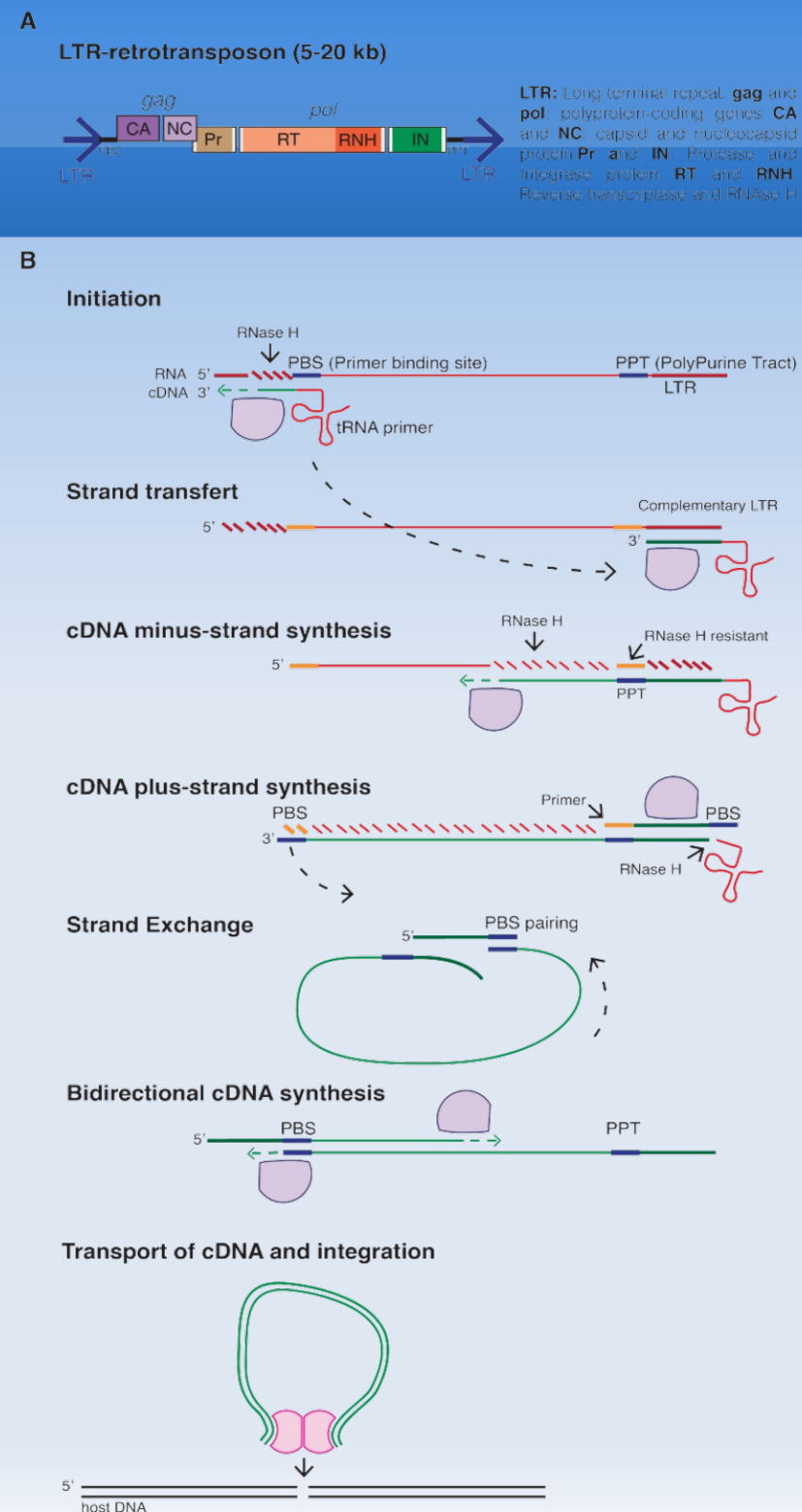
Genomic DNA that is transcribed to RNA and then reverse-transcribed back to DNA to be copied and pasted into different genomic locations

Long Terminal Repeat (LTR) retrotransposons ~8% of genome

Non-LTR regions: ~33%

- Long/Short Interspersed Nuclear Element:
- LINE (~7Kbp; 20%) encode proteins to "jump"
- SINE (~0.5Kbp; 13%) rely on LINEs to jump

Mutations have made most retrotransposons "dead" or dying, no longer capable of "moving"



Long interspersed Nuclear Element (LINE) retrotransposons

LINE retrotransposons make up about 21% of the human genome

We each have ~800,000 LINEs mostly non-functional

We may have ~100 functional retrotransposons, each ~7000 bp segments

LINE gene-complexes contain genes for reverse transcriptase, RNA-binding protein & nucleases but activity is regulated by host cell epigenetics and siRNA